

PHM_{turbofan}

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1 NASA C-MAPSS Turbofan Engine Dataset

The NASA C-MAPSS turbofan engine dataset is one of the most widely used benchmarks in prognostics and health management (PHM). It contains run-to-failure simulations of aircraft turbofan engines generated using the Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) platform developed at NASA Ames Research Center. The primary objective is to estimate the *Remaining Useful Life* (RUL) of an engine based on multivariate temporal measurements.

1.1 Dataset Description

Each engine is represented by a multivariate time series that records its operational behavior from an initially healthy state to a progressive degradation ending in system failure. The data include:

- an engine identifier,
- the operational cycle index,
- three operational settings that influence engine performance,
- twenty-one sensor measurements capturing physical parameters (e.g., temperatures, pressures, rotational speeds).

Each row corresponds to a single operational cycle, and each column corresponds to a specific variable. Engines begin with varying levels of initial wear and manufacturing variability, which are not considered faults but part of normal operating conditions. Sensor measurements are corrupted with realistic noise.

In the training sets, degradation progresses until failure. In the test sets, the trajectories terminate before failure, and the task is to predict the number of remaining cycles (RUL) after the last available measurement. Ground-truth RUL values are provided for all test trajectories.

1.2 Dataset Variants

The dataset is organized into four subsets (FD001–FD004), each representing different operational and fault scenarios:

- **FD001:** 100 training and 100 test engines; one operating condition; one fault mode (HPC degradation).
- **FD002:** 260 training and 259 test engines; six operating conditions; one fault mode.
- **FD003:** 100 training and 100 test engines; one operating condition; two fault modes (HPC and fan degradation).
- **FD004:** 248 training and 249 test engines; six operating conditions; two fault modes.

FD001 is the simplest scenario, whereas FD004 represents the most complex case due to multiple fault modes and multiple operational regimes.

1.3 Prediction Task

The goal of the prognostics task is to estimate the RUL for each engine in the test datasets. Formally, given a truncated multivariate time series $X = \{x_1, x_2, \dots, x_T\}$ representing the last T recorded operational cycles of an engine, the task is to predict the number of additional cycles the engine will operate before reaching a failure threshold.

2 Mathematical Modeling of the Time Series

The turbofan dataset consists of multivariate time series describing the temporal evolution of an engine’s operational state and degradation. Let each engine trajectory be represented as:

$$X = \{x_1, x_2, \dots, x_T\}, \quad x_t \in R^d,$$

where T is the number of recorded operational cycles and d is the number of measured variables (operational settings and sensor channels). Each observation vector can be decomposed as

$$x_t = [s_t, u_t],$$

where $s_t \in R^3$ denotes the operational settings and $u_t \in R^{21}$ the sensor measurements at cycle t .

2.1 State Degradation Process

The physical degradation of the engine is assumed to evolve as a latent, unobserved state variable z_t capturing the internal health condition of the system. A general stochastic state-space representation is

$$z_{t+1} = f(z_t, s_t) + \omega_t,$$

$$x_t = h(z_t, s_t) + \nu_t,$$

where f is the state-transition function, h relates the latent state to the sensor measurements, and ω_t, ν_t are process and measurement noise terms, respectively. The degradation process is monotonic in t and drives the system toward failure.

2.2 Remaining Useful Life Label

For each engine i in the training dataset, failure occurs at cycle F_i , and the Remaining Useful Life (RUL) at cycle t is defined as:

$$RUL_i(t) = F_i - t, \quad t = 1, \dots, F_i.$$

In the test dataset, the final cycle T_i is observed but F_i is unknown. The task is to predict

$$\widehat{RUL}_i = F_i - T_i.$$

2.3 Learning Objective

Given a learning model $\mathcal{M}$ parametrized by θ , the goal is to approximate the nonlinear mapping from the observed temporal sequence to its corresponding RUL:

$$\widehat{RUL} = \mathcal{M}(X; \theta).$$

The parameters θ are estimated by minimizing a loss function that penalizes the deviation between predicted and true RUL values. A commonly used choice is the mean squared error (MSE):

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N (\widehat{RUL}_i - RUL_i)^2.$$

2.4 Temporal Windowing

Because degradation is strongly correlated with recent temporal dynamics, models often operate on fixed-length windows extracted from the raw trajectories:

$$W_t^{(L)} = \{x_{t-L+1}, \dots, x_t\},$$

where L is the window length. The prediction target is the RUL associated with the last sample of the window. This sliding-window formulation transforms each sequence into a set of supervised instances, enabling the use of deep learning architectures such as CNNs, LSTMs, GRUs, and Transformers.

2.5 Assumptions

The modeling framework typically relies on the following assumptions:

- The degradation is progressive and irreversible.
- Sensor readings are noisy but informative proxies for the latent state.
- Engines of the same dataset follow similar degradation patterns under comparable operational conditions.
- The mapping from sensor data to RUL is smooth and learnable via non-linear function approximators.

This mathematical formulation provides a structured foundation for developing and evaluating prognostics models based on the NASA C-MAPSS turbofan engine dataset.

3 Reference

The dataset and simulation methodology are documented in:

A. Saxena, K. Goebel, D. Simon, and N. Eklund, “Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation,” *Proceedings of the 1st International Conference on Prognostics and Health Management (PHM08)*, Denver, CO, 2008.

The dataset is publicly available via the NASA Prognostics Center of Excellence.